Tennis Data Cleaning

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Introduction

This document contains the data collection and cleaning steps for our tennis project.

Our research question is:

Is there an association between the number of aces a player serves and winning a professional tennis match?

1. Data Collection

We use professional ATP tennis match data from 2023.

2. Data Cleaning

The Data set ("atp_matches_2023") contained 2,986 separate ATP professional matches and 49 variables. Some of the data points missed a variable. First I (Mickey) checked the data set for missing values of of our intended categories. I found 171 matches were missing ace statistics. We removed those matches because our research only focuses on the relationship between aces and winning. After removing the missing statistics we ended up with 2,815 matches to use as data points. We then selected nine variables relevant for the research (Player names, ace counts, court surface, match duration, and service points). Lastly we checked that there were no negative ace values and no negative values were found at all.

3. Exporting Cleaned Data

The cleaned dataset was exported as a CSV file named ("tennis_clean") containing 2,815 statistics and 9 variables which we will use for future analysis and visualizations.

I also separated the data cleaned data into 3 separate sub dataset each based on their own unique surface. The three are: - Hard - Grass - Clay

Since each surface effects the ball and the play style even slightly we believe it is much better to analyze each on its own and then compare to see if there are any meaningful differences.

```
tennis <- read.csv("Data/atp_matches_2023.csv")

head(tennis)
```

```
##   tourney_id tourney_name surface draw_size tourney_level tourney_date
## 1  2023-9900   United Cup    Hard        18             A     20230102
## 2  2023-9900   United Cup    Hard        18             A     20230102
## 3  2023-9900   United Cup    Hard        18             A     20230102
## 4  2023-9900   United Cup    Hard        18             A     20230102
## 5  2023-9900   United Cup    Hard        18             A     20230102
## 6  2023-9900   United Cup    Hard        18             A     20230102
##   match_num winner_id winner_seed winner_entry        winner_name winner_hand
## 1       300    126203           3                   Taylor Fritz           R
## 2       299    126207          NA                 Frances Tiafoe           R
## 3       296    126203           3                   Taylor Fritz           R
## 4       295    126207          NA                 Frances Tiafoe           R
## 5       292    126774           1              Stefanos Tsitsipas           R
## 6       291    207518          NA                 Lorenzo Musetti           R
##   winner_ht winner_ioc winner_age loser_id loser_seed loser_entry
## 1       196        USA       25.1   126610          5
## 2       188        USA       24.9   207518         NA
## 3       196        USA       25.1   128034          2
## 4       188        USA       24.9   200390         NA
## 5       193        GRE       24.3   126610          5
## 6       185        ITA       20.8   208852         NA
##              loser_name loser_hand loser_ht loser_ioc loser_age          score
## 1     Matteo Berrettini          R      196       ITA      26.7  7-6(4) 7-6(6)
## 2       Lorenzo Musetti          R      185       ITA      20.8    6-2 0-0 RET
## 3        Hubert Hurkacz          R      196       POL      25.8  7-6(5) 7-6(5)
## 4            Kacper Zuk          R      183       POL      23.9        6-3 6-3
## 5     Matteo Berrettini          R      196       ITA      26.7 4-6 7-6(2) 6-4
## 6 Stefanos Sakellaridis          R      196       GRE      18.3        6-1 6-1
##   best_of round minutes w_ace w_df w_svpt w_1stIn w_1stWon w_2ndWon w_SvGms
## 1       3     F     135    15    2     85      52       45       16      12
## 2       3     F      34     3    1     27      19       15        3       4
## 3       3    SF     104    11    0     80      50       44       19      12
## 4       3    SF      79     5    4     61      37       31       11       9
## 5       3    SF     155     9    1     98      52       43       28      16
## 6       3    SF      62     2    1     41      25       18       12       7
##   w_bpSaved w_bpFaced l_ace l_df l_svpt l_1stIn l_1stWon l_2ndWon l_SvGms
## 1         0         0     7    2     97      62       47       15      12
## 2         3         3     0    0     21      12        8        3       4
## 3         4         4    17    1     82      62       51        7      12
## 4         5         5     3    3     68      41       26       12       9
## 5         1         2     7    2     89      58       48       18      16
## 6         1         1     2    1     43      26       15        5       7
##   l_bpSaved l_bpFaced winner_rank winner_rank_points loser_rank
## 1         9         9           9               3355         16
## 2         1         3          19               2000         23
## 3         2         2           9               3355         10
## 4         6         9          19               2000        245
## 5         1         2           4               5550         16
## 6         4         9          23               1865        803
##   loser_rank_points
## 1              2375
## 2              1865
## 3              2905
## 4               220
```

```
## 5              2375
## 6                23
```

```
names(tennis)
```

```
##  [1] "tourney_id"         "tourney_name"       "surface"
##  [4] "draw_size"          "tourney_level"      "tourney_date"
##  [7] "match_num"          "winner_id"          "winner_seed"
## [10] "winner_entry"       "winner_name"        "winner_hand"
## [13] "winner_ht"          "winner_ioc"         "winner_age"
## [16] "loser_id"           "loser_seed"         "loser_entry"
## [19] "loser_name"         "loser_hand"         "loser_ht"
## [22] "loser_ioc"          "loser_age"          "score"
## [25] "best_of"            "round"              "minutes"
## [28] "w_ace"              "w_df"               "w_svpt"
## [31] "w_1stIn"            "w_1stWon"           "w_2ndWon"
## [34] "w_SvGms"            "w_bpSaved"          "w_bpFaced"
## [37] "l_ace"              "l_df"               "l_svpt"
## [40] "l_1stIn"            "l_1stWon"           "l_2ndWon"
## [43] "l_SvGms"            "l_bpSaved"          "l_bpFaced"
## [46] "winner_rank"        "winner_rank_points" "loser_rank"
## [49] "loser_rank_points"
```

```
colSums(is.na(tennis))
```

```
##         tourney_id       tourney_name            surface          draw_size
##                  0                  0                  0                  0
##      tourney_level      tourney_date          match_num          winner_id
##                  0                  0                  0                  0
##        winner_seed       winner_entry        winner_name        winner_hand
##               1736                  0                  0                  0
##          winner_ht         winner_ioc         winner_age          loser_id
##                 17                  0                  0                  0
##         loser_seed        loser_entry        loser_name        loser_hand
##               2217                  0                  0                  0
##           loser_ht         loser_ioc          loser_age              score
##                 32                  0                  2                  0
##            best_of              round            minutes              w_ace
##                  0                  0                232                171
##               w_df             w_svpt            w_1stIn           w_1stWon
##                171                171                171                171
##           w_2ndWon            w_SvGms          w_bpSaved          w_bpFaced
##                171                171                171                171
##              l_ace              l_df              l_svpt            l_1stIn
##                171                171                171                171
##           l_1stWon           l_2ndWon            l_SvGms          l_bpSaved
##                171                171                171                171
##          l_bpFaced        winner_rank winner_rank_points        loser_rank
##                171                 22                 22                 31
##  loser_rank_points
##                 31
```

```r
# Keep matches that have recorded ace statistics
tennis_clean <- subset(
  tennis,
  !is.na(w_ace) & !is.na(l_ace)
)

nrow(tennis_clean)
```

```
## [1] 2815
```

```r
tennis_clean <- tennis_clean[, c(
  "tourney_name",
  "surface",
  "winner_name",
  "loser_name",
  "w_ace",
  "l_ace",
  "minutes",
  "w_svpt",
  "l_svpt"
)]

head(tennis_clean)
```

```
##   tourney_name surface        winner_name            loser_name w_ace l_ace
## 1   United Cup    Hard       Taylor Fritz     Matteo Berrettini    15     7
## 2   United Cup    Hard     Frances Tiafoe       Lorenzo Musetti     3     0
## 3   United Cup    Hard       Taylor Fritz        Hubert Hurkacz    11    17
## 4   United Cup    Hard     Frances Tiafoe            Kacper Zuk     5     3
## 5   United Cup    Hard Stefanos Tsitsipas     Matteo Berrettini     9     7
## 6   United Cup    Hard    Lorenzo Musetti Stefanos Sakellaridis     2     2
##   minutes w_svpt l_svpt
## 1     135     85     97
## 2      34     27     21
## 3     104     80     82
## 4      79     61     68
## 5     155     98     89
## 6      62     41     43
```

```r
# Check for negative ace values
sum(tennis_clean$w_ace < 0)
```

```
## [1] 0
```

```r
sum(tennis_clean$l_ace < 0)
```

```
## [1] 0
```

```r
# Save the cleaned dataset
write.csv(
  tennis_clean,
```

```
  "Data/tennis_clean.csv",
  row.names = FALSE
)

# Different surfaces

tennis_grass <- subset(tennis_clean, surface == "Grass")
tennis_clay <- subset(tennis_clean, surface == "Clay")
tennis_hard <- subset(tennis_clean, surface == "Hard")

# Save

write.csv(tennis_grass, "Data/tennis_grass.csv", row.names = FALSE)
write.csv(tennis_clay, "Data/tennis_clay.csv", row.names = FALSE)
write.csv(tennis_hard, "Data/tennis_hard.csv", row.names = FALSE)
```

```
#testing the funciton

head(calculate_ace_difference(tennis_clean))
```

```
## [1]  8  3 -6  2  2  0
```